

JOAQUIN SIEBERT SANDOVAL

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EDUCATION

Columbia University

Master of Science in Data Science

New York, NY

Expected Dec 2026

University of Chile

Master's in Operations Management, GPA: 3.7/4.0

Bachelor of Science in Industrial Engineering, GPA: 4.0/4.0

Dean's List (Top 5%) 2019, 2020, and 2021. Relevant coursework: Linear algebra, probability, statistics, optimization.

Santiago, CL

Jan 2022

Jan 2022

HONORS & AWARDS

- Becas Chile Fellowship (2025) – Fully funded, merit-based national scholarship for graduate studies at Columbia University.
- Franz Edelman Award winner (INFORMS), 2022 – Project: “Analytics Saves Lives During the COVID-19 Crisis in Chile”.

PROFESSIONAL EXPERIENCE

Neuralworks SpA

Santiago, CL

Mid-Senior Data Scientist

Apr 2024 – Jul 2025

- Developed optimization models for aircraft engine maintenance using Gurobi-Python, decreasing operating costs by USD 2.6 million, due to avoiding leasing engines to cover anticipated operational unavailability for South America's largest airline.
- Designed and analyzed experiments (A/B tests) using R to evaluate operational improvements in aircraft maintenance systems.
- Mentored 4 mid-senior data scientists on career development and team practices using Scrum boards (Notion).

Tucar SpA

Santiago, CL

Head of Data

Jan 2023 - Apr 2024

- Led data integration and metrics optimization, setting a structured protocol to make the company's data 100% available for better decision-making across all teams.
- Created the company's data warehouse by orchestrating ETL pipelines (PySpark, Airflow) that migrated data from MongoDB to BigQuery, powering downstream analytics and BI reporting at scale.
- Built BI dashboards (Streamlit, Google Looker Studio) querying millions of rows to track customer behavior, and power heatmap/GPS cross-referencing analysis for fine assignment.

Data Scientist

Mar 2022 - Jan 2023

- Generated a Python (GeoPandas) hotspot-detection algorithm for impound and robbery events, reducing car thefts/inspections by 25% and operational costs by 6%.
- Developed and evaluated unsupervised learning pipelines (K-Means, hierarchical clustering) to improve customer risk segmentation and decision-making, reducing payment defaults by 70%.

RESEARCH EXPERIENCE

Institute of Complex Engineering Systems

Santiago, CL

Researcher

Sep 2020 - May 2022

- Crafted origin-destination models to forecast mobility flows in Santiago during COVID-19, supporting risk-zone labeling and public policy decisions.
- Constructed COVID-19 transmission simulation models in Python to assess Testing, Tracing, and Isolation policies for Chilean K-12 schools. Institutions that followed guidelines remained open for up to 80% of the time during 2020 and 2021.

TECHNICAL SKILLS

Languages & Libraries: Python (Pandas, Numpy, Sklearn, TensorFlow, SimPy, Gurobipy, LangChain, Transformers), R, SQL.

Data Platforms: GCP BigQuery, MongoDB, GCP Vertex AI.

Data Visualization: Power BI, Google Looker Studio, Streamlit.

Tools: GitHub, Jira, Confluence, Airflow, PySpark.

ADDITIONAL INFORMATION

Languages: Spanish (Native), English (Full professional proficiency).

Chilean citizen, Work authorization: OPT-eligible upon graduation (STEM extension eligible, up to 3 years). Also, eligible for H-1B1 (no lottery) for continued sponsorship.